CHAPTER 15

Closing Thoughts and Next Steps

Congratulations! You've made it to the end.

Let's take a moment to reflect on our journey throughout this book. We started with just the basics: what are containers and images? We saw how Docker provides packaging (images), delivery (automatic pulling of images), and an execution runtime (containers). We learned how these fundamental pieces provide a new way of thinking about software delivery.

The rest of the book has been an extended tutorial, guiding you through the process of creating, developing, and deploying a fully featured Rails app with Docker. We generated the Rails app using a container, and we created a custom image for running our Rails app, which we gradually enhanced. We introduced Docker Compose, using it to build up to a multiservice application with Redis, a database with decoupled volume for its data.

We made our Dockerized setup play nice with JavaScript, as we saw how to work with both the standard asset pipeline and `webpacker`, Rails' newest approach for integrating modern JavaScript front ends. We also covered how to test our app, going as far as running end-to-end tests relying on JavaScript in Chrome in both headless and non-headless mode.

However, even in Docker-land, nothing is perfect. We explored a couple of irritations you might encounter when working with Docker, and we did our best to mitigate these.

Finally, we started our journey toward production. After a quick tour of the production landscape to understand the capabilities we need and the tools available, we enhanced our app to be deployable in a production environment. We learned how to push our Docker images to a Registry, making them available for delivery to other machines. Next we introduced Swarm and used it to create our very own production-like playground on our local machine.